Design and Analysis of Algorithm (*CS3004-1*)

Multiple Choice Questions

Unit- 2

1` What is the primary idea behind the "Decrease and Conquer" technique?

a)Increase the problem size incrementally

b) Solve the entire problem in one step

c) Reduce the problem size at each step and solve it recursively

d) Divide the problem into multiple subproblems of equal size

2 Which of the following is a characteristic of the "Decrease and Conquer" method?

a) It always involves dividing the problem into two equal parts

b) It solves the problem by repeatedly reducing it to a smaller subproblem

c) It uses dynamic programming to store intermediate results

d) It is only applicable to sorting problems

3 In Insertion Sort, the problem size is reduced by:

a) Splitting the array into two parts of equal size

b) Sorting only the largest element in each iteration

c) Sorting a single element at a time

d) Exchanging adjacent elements

4 Which algorithm uses a variable-size decrease approach?

a) Exponentiation by squaring

b) Binary search

c) Euclid's algorithm

d) Insertion sort

5 Binary Search is a classic example of which variation of Decrease and Conquer?

a) Decrease by a constant

b) Decrease by a constant factor

c) Decrease by one

d) Divide and conquer

6 What type of data structure is used in Depth-First Search (DFS)?

a) Queue

b) Stack

c) Priority Queue

d) Heap

7 DFS is more suitable than BFS for which of the following?

a)Finding the shortest path in an unweighted graph

b) Exploring all vertices in a level-by-level manner

c) Detecting cycles in a graph

d)Solving shortest path problems in weighted graphs

8 What is the key difference between BFS and DFS in their approach?

a) BFS uses a stack; DFS uses a queue

b) BFS explores neighbors level by level; DFS explores as deep as possible along one branch

c) BFS works only on trees; DFS works only on graphs

d) BFS is recursive; DFS is iterative

9 Which of the following is true for BFS?

a) It always uses recursion

b) It can handle graphs with cycles by avoiding revisits using a "visited" array

c)It cannot be used to check graph connectivity

d) It is suitable only for weighted graphs

10 Which of the following problems can be solved using DFS?

a) Finding connected components in a graph

b)Finding the shortest path in an unweighted graph

c)Topological sorting of a directed acyclic graph (DAG)

d) Both a and c

11` **What is the primary application of topological sorting?**

a) Finding the shortest path in a graph

b) Sorting a list of elements in ascending order

c) Ordering the vertices of a directed acyclic graph (DAG) such that for every edge (u,v), vertex u comes before vertex v

d) Finding the minimum spanning tree of a graph

12 **Which of the following is a necessary condition for a graph to have a topological sorting?**

a) The graph must be a tree

b) The graph must be a DAG

c) The graph must be a complete graph

d) The graph must be a bipartite graph

13 **What is the time complexity of the topological sorting algorithm using Kahn's algorithm?**

a) O(V + E)

b)O(V * E)

c) O(E^2)

d) O(V^2)

14 **Which of the following is a correct statement about topological sorting?**

a) Topological sorting is only possible for graphs with a single source vertex

b) Topological sorting is only possible for graphs with a single sink vertex

c)Topological sorting can be performed on any DAG

d) Topological sorting can only be performed on graphs with a single cycle

15 **What happens if a graph has a cycle and we try to perform topological sorting on it?**

a)The algorithm will produce a correct topological sorting

b) The algorithm will produce an incorrect topological sorting

c) The algorithm will detect the cycle and report an error

d) The algorithm will go into an infinite loop

16 **Which of the following is an example of the Transform and Conquer method?**

a)Merge Sort

b) Binary search

c) matrix multiplication

d) Heap sort

17 **What operation is performed on an AVL tree when the balance factor of a node becomes greater than 1 or less than -1?**

a) Rotation

b) Deletion

c) Insertion

d) Search

18 **What is the condition for a binary search tree to be considered an AVL tree?**

a) The height of the left subtree and the right subtree of every node differs by at most 2

b) The height of the left subtree and the right subtree of every node differs by at most 1

c) The height of the left subtree and the right subtree of every node is always equal

d) The height of the left subtree and the right subtree of every node differs by at least 1

19 **What is the time complexity of search, insert, and delete operations in a 2-3 tree?**

a) O(log n)

b) O(n)

c) O(nlogn)

d) O(2^n)

20 **What happens when a node in a 2-3 tree is split due to the insertion of a new key?**

a) The node is replaced by two new nodes, each with one key

b) The node is replaced by three new nodes, each with one key

c) The node is not split, and the new key is added to the existing node

d) The node is replaced by two new nodes, each with one or two keys

21 **In a max-heap, which of the following is true?**

a) The parent node is smaller than its child nodes

b)The parent node is greater than or equal to its child nodes

c) The tree is a binary search tree

d) The elements are stored in ascending order

22 **What is the time complexity to build and heapify a heap of n elements?**

a) O(n log n)

b) O(log n)

c) O(n)

d) $O(n^2)$

23 **You are given an array: [3, 9, 2, 1, 4, 8]. After building a max-heap from this array, what will be the root element?**

a) 1

b) 9

c) 2

d) 8

24 **Which of the following applications is a heap NOT suited for?**

a) Implementing a priority queue

b) Finding the median of a data stream

c) Sorting data in ascending order

d) Depth-first search of a graph

25 **Which of the following is NOT true about heap sort?**

a) It is an in-place sorting algorithm

b) It is a stable sorting algorithm

c) It has a time complexity of O(n log n)

d) It uses the heap data structure

26 **What makes Counting Sort a distribution-based algorithm?**

a) It uses element frequencies to determine sorted positions.

b) It divides the array into smaller subarrays.

c) It recursively splits and merges data.

d) It compares each pair of elements to sort the array.

27 **Which sorting algorithm explicitly trades off space for time by using extra space to count occurrences of elements?**

a) Merge Sort

b) Bubble sort

c) Sorting by counting

d) Quick sort

28 **What is the time complexity of distribution counting sort?**

a) $\Theta(n^2)$

b) Θ(n)

c) O(n)

d) $O(n^2)$

29 **What is the time complexity of comparison counting sort?**

a) $\Theta(n^2)$ **b)** O(n)
c) $O(n^2)$ **d)** $\Theta(n)$

30 **Sorting by counting trades off space for time efficiency by:**
a) Using recursion to divide the array into smaller parts. **b) Allocating extra memory for the frequency and cumulative arrays.**
c) Avoiding comparisons between elements. **d)** Reducing the overall size of the input array.

31` **In Horspool's algorithm, when the last character of the pattern does not match the text, what action does the algorithm take?**
a) Shift the pattern by 1 character to the right b) Perform a full re-check from the beginning of the pattern
c) Shift the pattern by the value stored for the text character in the shift table d)Abort the search

32 **In Horspool's algorithm, when does the algorithm perform a full comparison of the pattern with the text?**
a) Every time a mismatch occurs b) Only when a match is found
c) Only when the shift table suggests no shift d) Initially, when the first character in the pattern is compared with the first character of the text

33 **In Horspool's algorithm, what is the primary purpose of the shift table?**
a) To store the indices of pattern matches b) To store the positions of all characters in the pattern
c) To store the number of shifts required for each character in the pattern d) To store the hash values of pattern characters

34 **In Horspool's algorithm, which table is used to store the shift values for the pattern matching process?**
a) Shift Table b) Hash Table
c) Match Table d) Character Table

35 **What is the primary advantage of Horspool's algorithm over the naive string matching algorithm?**
a) It guarantees finding all occurrences in linear time. **b) It uses a precomputed shift table to skip unnecessary comparisons.**
c) It always matches patterns in fewer comparisons than the naive algorithm. d) It works in constant space

36 **Which of the following is an example of a problem that can be solved using dynamic programming?**
a) Binary search **b) Knapsack problem**
c)Merge sort d)Depth First Search

37 **In dynamic programming, the solution to a problem is built up from:**

a) A single recursive call
b) A series of overlapping subproblems
c) A greedy approach
d) A divide and conquer method

38 **When solving a problem using dynamic programming, what is the role of "optimal substructure"?**

a) It ensures that solving subproblems will always result in the correct solution for the entire problem.
b) It guarantees that the problem can be solved in constant time.
c) It allows for efficient recursion without overlapping subproblems.
d)It simplifies the problem by eliminating unnecessary subproblems.

39 **What is the time complexity of solving the 0/1 knapsack problem using dynamic programming with an array-based approach?**

a) O(n^2)
b) O(n * W) where n is the number of items and W is the maximum weight
c) O(W)
d) O(n)

40 **What does the term "overlapping subproblems" refer to in dynamic programming?**

a) Problems that can be solved using greedy algorithms
b) Subproblems that appear multiple times in the recursive solution
c) Subproblems that are completely independent
d) Subproblems that cannot be solved independently

41` **What is the primary purpose of the Floyd-Warshall algorithm?**

a) Finding the shortest path between two specific nodes
b) Finding the shortest paths between all pairs of nodes in a graph
c) Finding the minimum spanning tree of a graph
d) Finding the longest path in a weighted graph

42 **Which type of graph is required for the Floyd-Warshall algorithm?**

a) Directed or undirected weighted graphs
b) Only directed weighted graphs
c) Only undirected graphs
d) Only acyclic graphs

43 **How does the Floyd-Warshall algorithm handle negative weight cycles?**

a) It detects them and stops execution
b) It ignores them and continues finding shortest paths
c) It gives incorrect results
d) It runs indefinitely

44 **Which technique is used to solve the 0/1 Knapsack problem efficiently?**

a) Greedy algorithm
b) Divide and conquer
c) Dynamic programming
d) Depth-first search

45 **What does the Floyd-Warshall algorithm store in its intermediate matrix?**

a) Edge weights only
b) Shortest path distances
c) Number of edges in the shortest path
d) The longest path in the graph

46 **Which of the following is a major limitation of the Floyd-Warshall algorithm?**

a) Cannot handle negative weights
b) High space complexity
c) Cannot find shortest paths for undirected graphs
d) Cannot find shortest paths for large graphs efficiently

47 **What is the space complexity of the Floyd-Warshall algorithm?**

a) O(V)
b) O(V^2)
c) O(V^3)
d) O(V log V)

48 **What is the Knapsack problem primarily used for?**

a) Finding the shortest path in a graph.
b) Maximizing the value of items in a knapsack without exceeding its capacity.
c) Sorting a list of items.
d) Finding the minimum spanning tree.

49 **Which of the following is true about memory function?**

a) It is a top-down approach
b) It is a bottom-up approach.
c) It does not use recursion.
d) It is only applicable to sorting algorithms.

50 **Which of the following problems can be solved using memory function?**

a) 0/1 Knapsack algorithms
b) Travelling salesman problem
c) Job assignment
d) All of the above.